

## 3.6 Modelling

The modelling phase highlights the core development of an optimization algorithm that is capable of generating the personalised and cost-effective Umrah trip plans. Previous phases show significant findings where Particle Swarm Optimization (PSO) is the selected option for this optimization mechanism that shows effectiveness in solving multi-objective and constrained criteria in defining optimization problems.

### 3.6.1 Problem Formulation

This research benchmark to achieve cost-effectiveness trips by implementing Particle Swarm Optimization (PSO) to search for the optimal combination. Each particle in the swarm represents a complete Umrah trip plan combination by constructing all the elements collected from the datasets. Preferences set up by the user-defined option to follow the suitable budget spending and travel duration for the whole optimization process. By iteratively updating particle positions and velocities, PSO is responsible for doing some exploration for the solution space and approaching cost-efficient solutions to satisfy the user's needs and constraints. This process of addressing the modelling of PSO in pseudocode can be seen in the **Table 3.5** below.

**Table 3.5**  
Modelling steps (Pseudocode)

| PSO Modelling Steps |                              |
|---------------------|------------------------------|
| Step 1              | Begin                        |
| Step 2              | Initialise PSO particles     |
| Step 3              | Evaluate fitness function    |
| Step 4              | Constraint handling          |
| Step 5              | Update velocity and position |
| Step 6              | Termination check            |
| Step 7              | Optimal solution             |
| Step 8              | End                          |

### 3.6.2 Design of PSO Particles Engine

In this research, a PSO particle is designed to allocate a candidate Umrah itinerary that consists of numerical parameters to represent the selected Umrah travel components, as example below:

- Return flight cost and number of stops
- Hotel cost in Makkah and Madinah
- Transportation cost
- Daily living and contingency expenses

Based on the listed travel components above, to present the realistic travel plan and logics based on users preference, the designated **Table 3.6** below will show the decision variables designed in this study.

**Table 3.6:**

Particle variables representation based on travel components

| Travel Components   | Particle Variables | Description                                                                                  |
|---------------------|--------------------|----------------------------------------------------------------------------------------------|
| Return Flight       | $F_i$              | Represents the index of a specific flight package.                                           |
| Makkah Hotel        | $H_i^M$            | Represents the index of a specific hotel in Makkah.                                          |
| Madinah Hotel       | $H_i^D$            | Represents the index of a specific hotel in Madinah.                                         |
| Transportation Tier | $T_i$              | Represents the service level (Low, Medium, or High) for inter-city and intra-city transport. |
| Daily Expenses Tier | $E_i$              | Represents the budget level (Minimal, Moderate, or Comfortable).                             |

By integrating the follows Umrah travel components into designated formula for the particle to present decision variable based on the following **Eq. (1)** below:

$$X_i = [ F_i, H_i^M, H_i^D, T_i, E_i ] \quad (1)$$

Where:

- $F_i$  = *return flight cost*
- $H_i^M$  = *hotel cost in Makkah*
- $H_i^D$  = *hotel cost in Madinah*
- $T_i$  = *transportation cost*
- $E_i$  = *daily expenses cost*

This representation gives a solid justification whereas the PSO algorithm enables to evaluate multiple travel combinations simultaneously and assess their feasibility based on user preference.

### 3.6.3 Fitness Function Design

The fitness function measures as the objective point used to evaluate and test each particle. In this study, the fitness function is designed to minimise the total Umrah trip cost. However, the cost calculation for these studies is unique and dynamic as the calculation will be based on user's specific input unlike static summation that will create the unlogic results. Accounting for the user's duration in Makkah ( $D_M$ ) and Madinah ( $D_D$ ).

These objective functions calculated as the sum of referred to the **Eq. (2)** below:

$$\min f(X_i) = C_{flight} + C_{accommodatiob} + C_{transport} + C_{dailyexp} \quad (2)$$

The above combination of fitness functions will be calculated as the whole budget that needs to be emphasized by the user during Umrah journey process. Considering travel components listed in the previous part in this study.

Moreover, the individual cost components are divided as follows cases:

### 1. Flight Cost ( $C_{flight}$ )

Calculated based on the selected flight price as in **Eq.3**

$$C_{flight} = Price(F_i) \quad (3)$$

### 2. Accommodation Cost ( $C_{accommodation}$ )

Calculated based on the selected hotel price in Makkah and Madinah rates multiplied by the total duration stays as in **Eq.4**

$$C_{accommodation} = (Rate(H_i^M) \times D_M) + (Rate(H_i^D) \times D_D) \quad (4)$$

### 3. Transportation Cost ( $C_{transport}$ )

Calculated based on the summation of fixed inter-city transfer costs and variable intra-city mobility costs based on transport tier ( $T_i$ ) as in **Eq.5**

$$C_{transport} = (C_{fixed}(T_i) + C_{internal}^M(T_i) \times D_M) + (C_{fixed}(T_i) + C_{internal}^D(T_i) \times D_D) \quad (5)$$

### 4. Daily Expenses Cost ( $C_{dailyexp}$ )

Calculated based on the selected expenses tier in ( $T_i$ ) multiplied by the total duration days as in **Eq.6**

$$C_{dailyexp} = Rate_{dailyexp}(E_i) \times (D_M + D_D) \quad (6)$$

This formulation that combines all the elements needed in Umrah planning, comprehend users to get a well-designated cost-effective Umrah plan based on the real-

world Umrah scenarios. However, penalty terms are embraced into the fitness function to settle down the constraint violations, such as when the user's budget is exceeded from the designated budget limit or flight options where it does not match with the user's preference for direct or transit flights.

### 3.6.4 Constraint Handling

Several constraints are applied to ensure solution feasibility will fit the budget constraint, flight type constraint and duration constraint that sum up all these criteria to create the penalty based fitness function as follows.

#### Budget constraint

That specifies the budget constraint matches with the user's input based on the number of budget trips as in **Eq. (7)**.

$$f(X_i) \leq B_{user} \quad (7)$$

Where:

- $B_{user} = \text{user defined budget}$

#### Flight type constraint

That specifies the trip duration matches with the user's input based on the type of flight user choose as in **Eq. (8)**.

$$S_{out} \leq S_{max}, \quad S_{return} \leq S_{max} \quad (8)$$

Where:

- $S_{out} = \text{number of outbound stops}$
- $S_{return} = \text{number of return stops}$
- $S_{max} = \text{evaluated as 0 for direct flights or } \geq 1 \text{ for transit flights}$

### Duration constraint

That specifies the trip duration matches with the user's input based on the number of day trips as in **Eq. (5)**.

$$D_{trip} = D_{user} \quad (9)$$

### Penalty based fitness function:

By handling constraint, a penalty based fitness function is used as below as in **Eq. (10)**:

$$f'(X_i) = f(X_i) + \lambda_1 P_{budget} + \lambda_2 P_{flight} \quad (10)$$

Where:

- $P_{budget} = \max(0, f(X_i) - B_{user})$
- $P_{flight} = \text{penalty if flight type violates preference}$
- $\lambda_1, \lambda_2$  are the penalty weights

### 3.6.5 PSO Velocity And Position Update

Based on the each particle assigned has updated its velocity and position using the formula as followed by the **Eq. (11)** and **(12)** below:

$$v_i^{t+1} = wv_i^t + c_1r_1(pbest_i - x_i^{t+1}) + c_2r_2(gbest_i - x_i^t) \quad (11)$$

$$x_i^{t+1} = x_i^t + v_i^{t+1} \quad (12)$$

Where:

- $w = \text{inertia weight}$
- $c_1, c_2 = \text{cognitive and social coefficients}$
- $r_1, r_2 = \text{random numbers in } [0,1]$

### 3.6.6 Termination Check

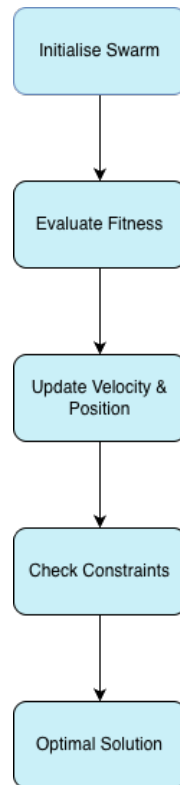
The PSO optimization process is iterative and the algorithm is designed to refine the candidate solutions over multiple cycles. To prevent infinite loops and ensure the computational process is efficient, a termination process check is designated to encounter those problems. In this research, the algorithm is set to the limit of 100 iterations and automatically terminated if the number of iteration ( $T_{max}$ ) has achieved the maximum numbers.

Each iteration that ended is checked by the algorithm to see either the current iteration count ( $t$ ) = ( $T_{max}$ ) or not. If the iteration count ( $t$ ) < ( $T_{max}$ ), the algorithm will loop back to step 2 of evaluating fitness function to keep refine the velocity and positions of the swarm.

### 3.6.7 Optimal Solutions

After assigning all the optimization process from initializing the PSO particles as measured variables proceeding to fitness function design to evaluate each particle. Moreover, considering the constraint handling penalty based on the fitness function. Once the termination criteria is met, the entire search process will identifies the Global Best Particle ( $G_{best}$ ) based on the recorded data by the algorithm.

To process all the scenarios cases that being developed for the optimization process can be referred as below figure where PSO based optimization enquiry these steps as shown in **Figure 3.10** below:



**Figure 3.10:** Optimization Process Steps

This modelling approach suits with the objective of making an effective approach where it captures the multi-dimensional nature of umrah trip planning. Integration with real travel data that combined with a PSO-based optimization architecture to efficiently use solutions to make searching space and generate realistic budget costs that minimise the travel plans cost that align with user preference.